MIL-E-36607
13 October 1969

MILITARY SPECIFICATION

EXTRACTOR-IMPACTOR-DRIVER, HIP BONE NAIL, JEWETT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an extractor-impactor-driver, suitable for use with hip nail and bone plates of the Jewett design, used in the orthopedic repair and reconstruction of the hip.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (for sealing fiberboard boxes).
GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for carton sealing).

FSC 6515

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Military

MIL-P-116 Preservation, Methods of.

STANDARDS

Federal

Test Method 151 Metals; Test Methods.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24192 - Extractor-Impactor-Driver, Hip Bone Nail, Jewett

(Copies of specifications, standards, and drawing, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

UNITED STATES OF AMERICA STANDARDS INSTITUTE (U.S.A.S.I.)
U.S.A.S.I. B1.1 Unified Screw Threads
U.S.A.S.I. B1.2 Gages and Gaging for Unified Screw Threads

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016).

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3. REQUIREMENTS

3.1 Material. The entire extractor-impactor-driver assembly shall be fabricated from Class 3, type 302, 303 or 304 material in accordance with Federal Specification GG-I-526.

3.2 Style, design and dimensions. Shall be a hip nail and bone plate extractor-impactor-driver of the Jewett design. Shall be suitable for extracting, driving and impacting of hip nail and bone plate combinations of the Jewett design, used in the orthopedic repair and reconstruction of the hip. The assembly shall be cannulated throughout its length to completely pass over Steinmann bone pins and Kirschner bone wire drills of 0.0938 inch diameter. The external thread of the assembly used in extracting Jewett nail and bone plate combinations shall be in strict accordance with U.S.A.S.I. Standard B1.1 for 1/4-20 UNC-2A (Unified Screw Thread System of Dimensions). The entire assembly shall be in strict accordance with DPSC Drawing 24192.

3.3 External threads of extractor. The external threads of the extractor which engage the Jewett hip nails shall allow the "GO" thread ring gages to pass over their full length and turn freely and shall not allow passage of the "LO" and "NOT GO" thread ring gages when tested as specified in 4.4.2. Definitions of acceptance and rejection of the gaging tests shall be as specified in U.S.A.S.I. Standard B1.2.

3.4 Hardness. Rockwell hardness of the barrel, knob and adjusting nut of extractor shall be B 75 minimum after proper stress relieving when tested as specified in 4.4.3.

3.5 Finish. The finished assembly shall have a smooth mirror finish, or mirror and satin finish, properly passivated and electropolished.

3.6 Resistance to corrosion. The finished assembly shall show resistance to the boil test as specified in 4.4.1.

3.7 Identification markings. The extractor shall be permanently and legibly marked with the name or registered trademark of the contractor and an indication that material is stainless steel. Extractors of foreign (other than U.S.) manufacture shall indicate, in addition to the above, the country of origin.

3.8 Workmanship. Workmanship shall be first class throughout. Extractors shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I.

Table I. Sampling Criteria for Examination

Type Defect	Inspection Level *	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5

* If acceptance number is not exceeded, lot shall be accepted. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

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4.2.2 For Tests. Sampling for tests shall be conducted in accordance ith MIL-STD-105, and as indicated in Table II.

Table II. Sampling Criteria for Tests

omponent	Characteristic	Req't	Test Procedure	Unit of Product	Insp. Level	AQL
xtractor -	Dimensions	3.2	-------	1-extractor	S-2	*
mpactor -	External Thread Test	3.3	4.4.2	"	S-4	*
river	Hardness	3.4	4.4.3	"	S-2	*
	Boil Test	3.6	4.4.1	"	S-2	*

Acceptance number shall be zero for all sample sizes

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the onstruction of the extractors shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be sed for the determination of two or more test characteristics.

4.3 Examination. The extractor-impactor-drivers shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables I and III and in accordance with IL-STD-105. Inspection, however, shall not be restricted to defects listed in able III. Any extractor-impactor-drivers in the sample containing one or more lefects shall be rejected and if the number of defective items in any sample xceeds the acceptance number for that sample, the lot represented by the sample hall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical:	None defined
Major	
101	Extractor assembly not of finish specified.
102	Extractor barrel not free of sharp edges.

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Table III. Classification of Defects (Cont'd)

Categories	Defects
Major	
103	External thread member of extractor, 1/4-20 UNC-2A, not permanently secured to body.
104	External thread member of extractor, 1/4-20 UNC-2A, not complete or well defined.
105	Assembly not able to completely pass over bone pins and wire drills of 0.0938 inch diameter.
Minor	
201	Extractor assembly not identified as specified in 3.7.

4.4 Tests. Tests shall be conducted to determine compliance with spe fication requirements.

4.4.1 Corrosion test. The finished assembly shall be subjected to th boil test as specified in GG-I-526 for class 3 material.

4.4.2 External thread test. The contractor shall provide all of the lowing thread ring gages to the Government Quality Assurance Representative (for checking the 1/4-20 UNC-2A external threads of the extractor with respect Pitch Diameter and Major Diameter.

a. "GO" Thread Ring Gage for PITCH DIAMETER (X gage) Identificatio marking: 1/4-20 UNC 2A GO PD .2164
b. "LO" Thread Ring Gage for PITCH DIAMETER (X gage) Identificatio marking: 1/4-20 UNC 2A LO PD .2127
c. "GO" Plain Ring Gage for MAJOR DIAMETER (Z gage) Identification marking: GO .2489
d. "NOT GO" Plain Ring Gage for MAJOR DIAMETER (Z gage) Identific marking: NOT GO .2408

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The gaging procedure to be used, dimensions, tolerances and gage identifications shall be in accordance with U.S.A.S.I. Standard B1.2. Fits of gages onto external extractor threads shall be noted.

4.4.3 Hardness test. Rockwell hardness shall be determined in accordance with method 243 of Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One extractor-impactor, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	48 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-15.

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5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type: PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 This specification is intended to cover extractor-impactor-driver, hip bone nails which are normally procured to meet Military requirements. It does not include all types, sizes, etc., as indicated by the title, nor all types which are commercially available.

6.2 This specification covers the following item listed in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-462-0834	EXTRACTOR-IMPACTOR-DRIVER, HIP BONE NAIL, Jewett

6.3 Shall be a hip nail extractor-impactor-driver of Jewett design suitable for use with the nail and bone plate family of items under Federal Stock numbers as follows:

FSNs
6515-145-0139
6515-145-0141
6515-462-0841
6515-462-0844
6515-462-0845
6515-462-0846

FSNs
6515-464-2522
6515-464-2529
6515-464-2530
6515-464-2544
6515-464-2550

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION
MIL-E-36607 FSN

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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